



What is DevOps and Its Principles

Definition, Principles, and Benefits | DevOps Culture | Key Practices

What is DevOps?

Definition

DevOps combines **Development (Dev)** and **Operations (Ops)**.

Goal

- Faster software delivery
- Improved collaboration
- Higher software quality

Why DevOps?

- 1 Reduces SDLC time**
- 2 Enables faster releases**
- 3 Increases team collaboration**
- 4 Enhances system reliability**



DevOps Principles



Collaboration



Automation



**Continuous
Improvement**



Security



Benefits of DevOps

Faster Deployment

Improved Quality

Efficient Resolution

Scalability & Flexibility

Cost Reduction

DevOps Culture

1

Shift Left Mindset

Early testing & security

2

Blameless Postmortem

Learn from failures

3

Continuous Learning

Encourage skill enhancement

4

Customer-Centric Approach

Prioritize user feedback





Continuous Integration
or delivery delivery



Infrastructure as code
on monitrction connn feedback

Key DevOps Practices

1

CI/CD

2

IaC

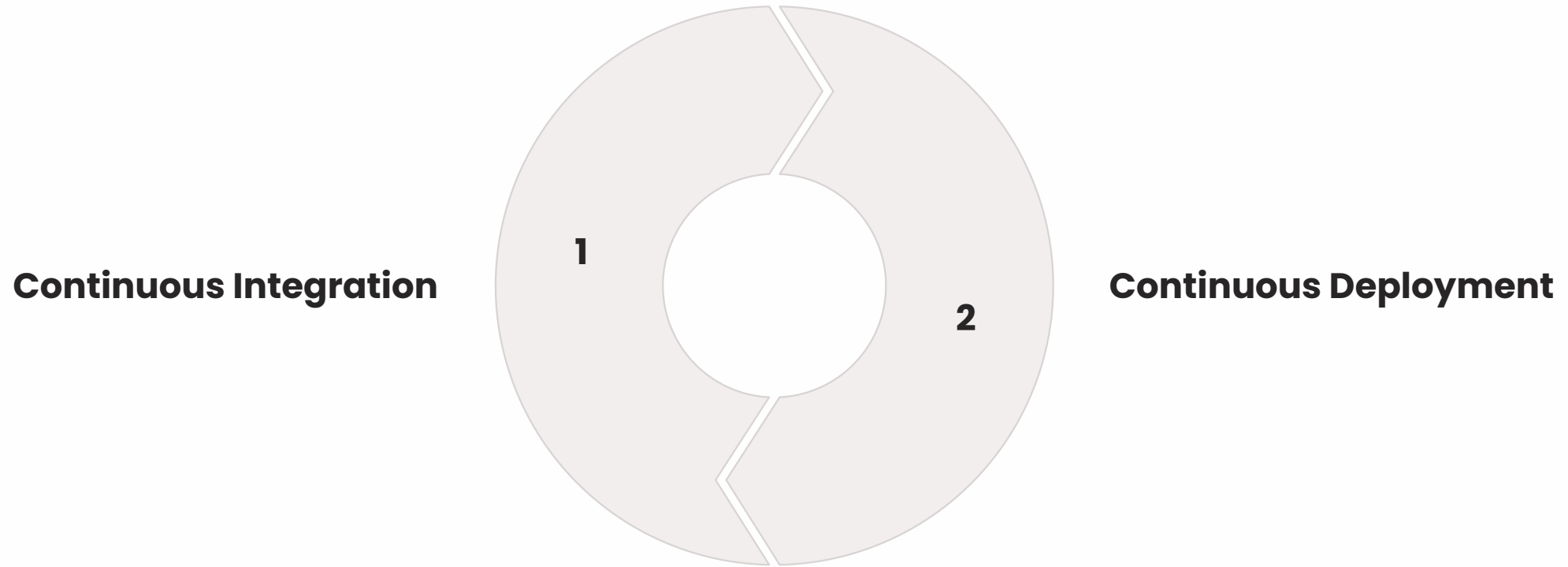
3

Automated Testing

4

**Monitoring &
Logging**

CI/CD



Automating code integration & testing. Automatically deploying updates to production.

Infrastructure as Code (IaC)

1

Scalability

2

Consistency

3

Faster Deployments

Managing infrastructure using code. Tools: Terraform, Ansible, CloudFormation



Automated Testing

Unit Testing

Integration Testing

Functional Testing

Performance Testing

Tools: Selenium, JUnit, PyTest

Monitoring & Logging

100%

System Performance

99.9%

Reliability

24/7

Quick Issue Detection

Ensures system performance & reliability. Tools: Prometheus, Grafana, ELK Stack. Benefits: Quick issue detection, reduced downtime



Conclusion

- DevOps transforms development
- Automate, collaborate, improve
- Keep learning & adapting

